

An AI-Enhanced Framework for Smart General Clinic Management: Digital Transformation in Vietnam’s Healthcare

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Abstract

Abstract

As digital transformation reshapes the healthcare sector, optimizing operational workflows in general clinics has become a critical necessity [2]. This paper presents the design of a Smart Hospital Management System (SHMS) that integrates Artificial Intelligence (AI) to digitize the entire medical process, including reception, scheduling, Electronic Medical Records (EMR), and billing [1]. A key innovation of this system is the application of AI for preliminary symptom analysis and prescription support, which significantly reduces the administrative burden on medical staff and minimizes patient waiting times [3]. By unifying actors—patients, clinicians, and administrative staff—into a cohesive digital ecosystem, the system leverages Role-Based Access Control (RBAC) and AI-driven symptom analysis to ensure clinical efficiency and data security [4, 5]. The proposed framework aims to enhance user experience and improve healthcare governance efficiency within the Vietnamese medical context [6].

1 Introduction

The global healthcare industry is undergoing a profound digital transformation. Despite this, many hospitals and clinics in Vietnam still rely on manual processes characterized by fragmented paper records, prolonged waiting times, and a lack of continuous connection between

patients and doctors [1]. There is a surging demand for real-time appointment booking, on-line health monitoring, and immediate digital access to medical results to eliminate redundant hospital visits [1, 2].

Our proposed Smart Hospital Management System (SHMS) is an integrated platform designed to digitize the end-to-end clinical workflow: from patient registration and scheduling to EMR updates, safe prescribing, and automated billing [2]. Unlike traditional management software, this system incorporates advanced Role-Based Access Control (RBAC) for data security and utilizes AI modules to assist in clinical diagnosis, providing a path toward predictive and personalized medicine [3, 4]. The primary goal is to optimize the healthcare delivery process, shorten waiting periods, and improve the overall efficiency of medical professionals [2].

2 Related Work

Current digital health platforms such as **Practo** and **Hello Bacs**i have established significant footprints in the Southeast Asian market [7]. Practo provides a comprehensive network for booking and personal health record management, but it is not fully optimized for the localized insurance workflows in Vietnam and lacks deep AI-integrated diagnostic support for clinicians [8, 9]. Hello Bacs*i* offers a rich content repository and basic health consultation chatbots but remains limited in internal hospital operations, such as clinic room distribution, queue management, and complex multi-role permission systems [5, 9].

Modern "Smart Hospital" research focuses on the synergy between the Internet of Medical Things (IoMT), Big Data, and AI to personalize patient care [10, 11]. Recent studies highlight that AI-driven Clinical Decision Support Systems (CDSS) can significantly reduce medication errors, which are a major cause of mortality worldwide [12, 13]. Furthermore, advanced algorithms such as Deep Reinforcement Learning are being utilized to solve complex outpatient scheduling problems, reducing the cumulative duration of patient engagement within the facility [14]. Our system builds upon these foundations by integrating these functionalities into a single, localized platform [5].

3 Proposed Methodology

The system is designed using an object-oriented approach involving five primary actors: Patient, Doctor, Receptionist, Administrator, and AI Service [6, 15].

3.1 System Actors and Functional Scope

The SHMS serves four main user groups with specialized functionalities [16]:

- **Patient:** Can perform account authentication via email or VNeID, manage digital health insurance, use personal QR codes for rapid check-in, and track appointment status in real-time [15, 17].
- **Doctor:** Manages consultation schedules, updates EMRs, uploads lab results, and receives AI-driven suggestions for preliminary diagnosis and safe drug dosages [18, 19].
- **Receptionist:** Handles patient intake, confirms schedules via QR codes, manages billing, and coordinates clinic room assignments and queues [19, 20].

- **Administrator:** Controls system-wide configurations, manages user roles through RBAC, and monitors the "Audit Log" to track administrative actions [4, 19].

3.2 AI Service Integration

The system integrates an "AI Service" actor to provide three intelligent modules [4, 7]:

1. **AI Chatbot:** Provides 24/7 automated support for FAQs and appointment guidance [4].
2. **Smart Recommendation:** Automatically suggests appropriate clinical departments and specialist doctors based on patient-entered symptoms [7, 21].
3. **Health Analysis:** Analyzes historical medical data to alert clinicians of health risks and provide preliminary diagnostic candidates [7, 21].

4 Conclusion

This paper presented a framework for an AI-powered Smart Clinic Management System. By consolidating the needs of all hospital actors onto a single platform, the solution addresses the limitations of existing systems like lack of diagnostic support and fragmented operations [5]. Future work will focus on integrating IoMT devices for continuous patient monitoring and refining AI algorithms using local clinical datasets to enhance diagnostic accuracy.

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